

2.3 Assess Algorithm Efficiency

کارکردگی (Efficiency) وقت (رفتار) اور جگہ (میموری) سے ناپی جاتی ہے۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 08 of 24

2.3 Assess Algorithm Efficiency

LEARN IT

Algorithm efficiency is measured in terms of **time** and **space** — how fast it runs and how much memory it uses.

The key considerations for assessing efficiency are:

1. **Number of steps** (described with **Big O notation**).
2. **Number of conditions** (IF decisions).
3. **Number of repetitions** (loops).

Time (speed) + Space (memory) → Efficiency

WORKED EXAMPLE

Worked example: two ways to find a name in 1,000,000 sorted records.

Linear search checks about 1,000,000 names in the worst case (time grows with n). Binary search checks about 20 names (it halves the list each time). Both can be correct and clear. They are not equally efficient.

REMEMBER IT

Efficiency is always “as the input grows”. An algorithm that feels instant on 10 items can freeze on 10 million items if it is $O(n^2)$.

COMMON MISTAKE

Mistake: judging efficiency by how short the code looks.

Fix: a five-line nested loop can do n^2 work. Count steps, conditions and repetitions — not lines of text.

اردو میں سمجھیں

کارکردگی (Efficiency) وقت (رفتار) اور جگہ (میموری) سے ناپی جاتی ہے۔ تین چیزیں دیکھی جاتی ہیں: قدموں کی تعداد (بگ او)، شرطوں کی تعداد، اور دہرائی (لوپ) کی تعداد۔ چھوٹے ان پٹ پر تیز لگنے والا الگورتھم بڑے ان پٹ پر سست پڑ سکتا ہے۔

CHECK YOURSELF

1. Efficiency is measured in terms of which two resources?
2. List the three key considerations named in the source.

Answers: 1. Time (speed) and space (memory). 2. Number of steps (Big O), number of conditions, number of repetitions.

Next lecture: 2.3.1 Number of Steps (Big O Notation)